

Machine Learning for DRG Prediction: A Neural Network Approach

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ABSTRACT The classification of patients into Diagnosis-Related Groups (DRGs) is a critical task in hospital management and health economics. DRGs categorize cases into groups that are clinically similar and are expected to consume similar amounts of hospital resources. Currently, DRG assignment is a complex process performed by medical coders. This process is time-consuming, prone to human error, and requires highly specialized personnel. This study addresses the problem of predicting the corresponding DRG for a patient based on unstructured arrays of diagnoses, procedures, and basic demographic data (age and sex) from Hospital El Pino. By leveraging Machine Learning (ML) techniques—including tree-based ensemble methods and Multilayer Perceptrons (MLPs)—we aim to automate and assist in the DRG classification process across 526 unique categories. The best model achieved a Top-5 Accuracy of 86.4% and a Weighted F1-Score of 0.61 on the held-out test set. A domain-specific chatbot (KONOHA-BOT) was additionally developed using prompt engineering and a local LLM as part of Phase 2.

INDEX TERMS Diagnosis-Related Groups (DRG), Artificial Neural Networks, Hospital Management, Machine Learning, Predictive Modeling, Prompt Engineering, Chatbot.

I. INTRODUCTION

THE classification of patients into Diagnosis-Related Groups (DRGs) is a critical task in hospital management and health economics. DRGs categorize cases into groups that are clinically similar and are expected to consume similar amounts of hospital resources. Currently, DRG assignment is a complex process performed by medical coders who analyze patient records, including principal and secondary diagnoses, as well as performed procedures.

This process is time-consuming, prone to human error, and requires highly specialized personnel. By leveraging Machine Learning (ML) techniques we can automate and assist in the DRG classification process. This study addresses the problem of predicting the corresponding DRG for a patient based on unstructured arrays of diagnoses, procedures, and basic demographic data (age and sex) from Hospital El Pino.

II. RELATED WORK

Previous studies have tackled the problem of DRG prediction using various computational approaches. Cheng et al. [1] focus on the sequentiality of diagnoses using recurrent networks, while Kusunoki et al. [2] prioritize natural language

processing to extract meaning from clinical notes. Pérez and Gómez [3] investigate tabular ML models such as Random Forests and Gradient Boosting combined with demographic data.

Our project differs by proposing a pipeline that integrates TF-IDF vectorization of coded clinical data (ICD-10) with both classical ensemble methods (Random Forest, XGBoost) and deep feedforward networks (MLP), seeking a balance between computational complexity and accuracy in the multiclass classification of 526 DRG categories.

III. OBJECTIVE OF THE STUDY

The main objective of this study is to develop and evaluate Machine Learning models capable of accurately predicting a patient's Diagnosis-Related Group (DRG) using their age, sex, and lists of primary and secondary diagnoses and procedures, and to compare classical ensemble approaches against deep learning architectures on this task.

IV. METHODOLOGY

A. DATASET DESCRIPTION

The dataset corresponds to patient discharge records obtained from Hospital El Pino. It consists of 14,561 records and 68 columns. The features include:

- **Diagnoses (35 columns):** 1 principal diagnosis and up to 34 secondary diagnoses (CIE-10 format, code + description).
- **Procedures (30 columns):** 1 principal procedure and up to 29 secondary procedures.
- **Demographics:** Age in years (Edad en años) and Sex (Sexo).
- **Target Variable:** The assigned DRG (GRD), which encompasses 526 unique categories.

Due to the structure of the data, the dataset is highly dimensional and sparse, given that most patients do not have 35 diagnoses or 30 procedures.

B. METHOD

This study follows the Cross-Industry Standard Process for Data Mining (CRISP-DM) methodology, which provides a structured approach for data-driven projects. The process includes the following phases: Business Understanding, Data Understanding (EDA), Data Preparation, Modeling, Evaluation, and Deployment.

C. PROCESS FOR DEVELOPING THE MODEL

- **Data Preprocessing:** Handle missing values (represented as '-' in diagnosis and procedure columns). Extract medical codes and apply TF-IDF vectorization. Age is numerically scaled and Sex is one-hot encoded.
- **Feature Engineering:** Clinical text (diagnoses and procedures) is concatenated and transformed via `TfidfVectorizer` (1,000 features), capturing statistical importance of rare pathologies over generic terms.
- **Model Training:** The dataset is divided into Training (80%) and Testing (20%) subsets using stratified hold-out to evaluate model generalization.
- **Cost Function:** Cross-Entropy Loss for the MLP; F1-Macro as the optimization criterion for GridSearchCV on classical models.
- **Hyperparameter Tuning:** GridSearchCV with 3-fold cross-validation for tree-based models; manual grid search over hidden dimensions and learning rates for the MLP.

D. MACHINE LEARNING TECHNIQUES

Three architectures were evaluated:

- 1) **Random Forest (Baseline):** Traditional ensemble with `class_weight='balanced'` to handle class imbalance.
- 2) **XGBoost (GPU Accelerated):** Gradient boosted trees with `tree_method='hist'` and CUDA acceleration, with sample weights inversely proportional to class frequency.

- 3) **Deep MLP (PyTorch):** A dense feedforward network with two hidden layers, ReLU activations, Dropout (0.3), and Adam optimizer, trained with Early Stopping (patience = 8 epochs).

E. EVALUATION METRICS

Given 526 unique DRG classes and significant class imbalance, standard accuracy alone is insufficient. The following metrics were employed:

- **Macro F1-Score:** Penalizes the model if it ignores rare DRG classes.
- **Weighted F1-Score:** Measures overall effectiveness weighted by class frequency.
- **Top-5 Accuracy:** Evaluates whether the correct DRG appears among the top 5 model predictions, the most clinically relevant metric for assisting human coders.

V. EXPLORATORY DATA ANALYSIS

A. DATA QUALITY STUDY

1) Completeness

The dataset presents no missing values in the Edad en años, Sexo, and GRD columns. However, diagnosis and procedure columns are inherently sparse. Empty slots are filled with the symbol '-', which must be treated as missing data (NaN) during preprocessing.

2) Correctness & Outliers

The age variable ranges from 0 to 121 years. The upper bound, while unusual, remains clinically possible and was retained. No anomalous categories were found in the Sexo column (only Hombre and Mujer were observed).

B. DESCRIPTIVE STATISTICS

Table 1 presents an executive summary of the 14,561-record dataset. The distribution by sex reveals a clear predominance of female patients (65.9%), consistent with the hospital's obstetric focus. The target variable exhibits a severe long-tail distribution across 526 classes, with the top DRG (PH Cesárea) accounting for only 5.6% of cases.

TABLE 1. Descriptive Summary of Demographic and Clinical Variables from Hospital El Pino

Metric	Value
Total records	14,561
Total features	68
Age Mean	40.3 years
Age Range	0–121 years
Female	9,617 (65.9%)
Male	4,944 (34.1%)
Avg. diagnoses/patient	7.11
Avg. procedures/patient	13.22
Unique GRD classes	526
Top GRD	PH Cesárea (813 cases, 5.6%)

C. VISUALIZATIONS

Fig. 1 shows the age distribution, revealing a peak near age 0 (neonatal admissions) and a secondary peak between ages

20–35 (maternity). Fig. 2 illustrates the 65.9/34.1 sex split and confirms that female patients exhibit a lower median age, consistent with obstetric admissions. Fig. 3 demonstrates the extreme long-tail class imbalance of the target variable. Fig. 4 confirms that the top DRG categories are strongly associated with specific age ranges and are female-exclusive.

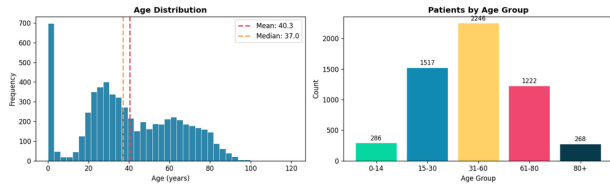


FIGURE 1. Histogram and bar chart of the age distribution and patient age groups.

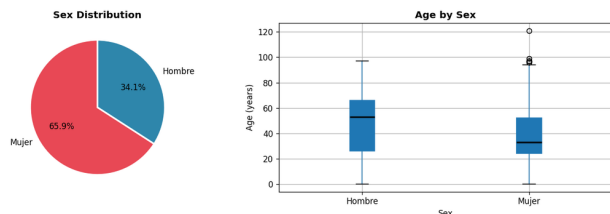


FIGURE 2. Pie chart of sex distribution and boxplot of age distribution by sex.

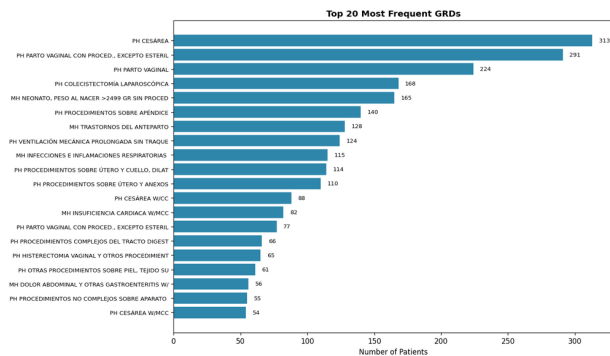


FIGURE 3. Bar chart of the top 20 most frequent GRDs, demonstrating severe class imbalance.

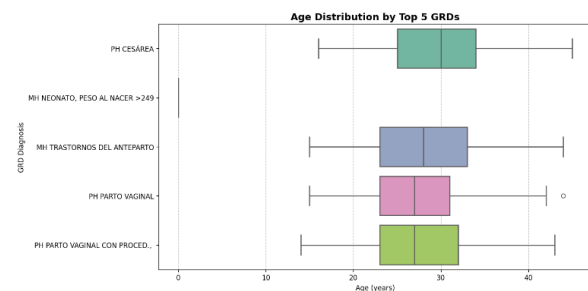


FIGURE 4. Boxplot of age distribution by the top 5 GRDs, showing obstetric DRGs are restricted to specific age ranges.

VI. EXPERIMENTS: RESULTS AND DISCUSSION

A. EXPERIMENTAL SETUP AND PREPROCESSING

To address DRG prediction, a robust ML pipeline was implemented to handle both free-text clinical variables and demographic data, while mitigating the severe class imbalance (526 classes, 38% with fewer than 5 records).

The dataset was split using stratified hold-out: 80% for training and cross-validation, 20% as a strictly isolated test set for final evaluation. A ColumnTransformer unified pre-processing across modalities:

- **Clinical Text:** Diagnoses and procedures were concatenated and transformed via TfidfVectorizer (1,000 features), producing a global sparse frequency matrix.
- **Numerical:** Age was standardized with StandardScaler.
- **Categorical:** Sex was encoded via OneHotEncoder.

B. CANDIDATE ARCHITECTURES AND HYPERPARAMETER OPTIMIZATION

Three model families were evaluated on the same feature space:

1) Random Forest (Baseline)

A classical ensemble with `class_weight='balanced'`. GridSearchCV explored `n_estimators` $\in \{50, 100\}$ and `max_depth` $\in \{20, \text{None}\}$ with 3-fold CV. Best configuration: `n_estimators=100`, `max_depth=None`.

2) XGBoost (GPU)

Given the high dimensionality of the TF-IDF matrix, XGBoost was trained with `tree_method='hist'` and `device='cuda'`. Sample weights inversely proportional to class frequency were applied to correct class imbalance in the gradient optimization. GridSearchCV explored `n_estimators` $\in \{50, 100, 200\}$ and `max_depth` $\in \{10, 20, 50\}$. Best configuration: `n_estimators=200`, `max_depth=10`.

3) Deep MLP (PyTorch)

A dense feedforward network with the architecture:

$$\text{Input} \rightarrow \text{FC1002} \rightarrow \text{ReLU} \rightarrow \text{Dropout}(0.3) \rightarrow \text{FC}_h \rightarrow \text{ReLU} \rightarrow \text{Dropout}(0.3) \rightarrow \text{FC526}$$

where $h \in \{128, 256\}$ and learning rate $\in \{0.001, 0.0001\}$ were explored under the Adam optimizer with CrossEntropy-Loss.

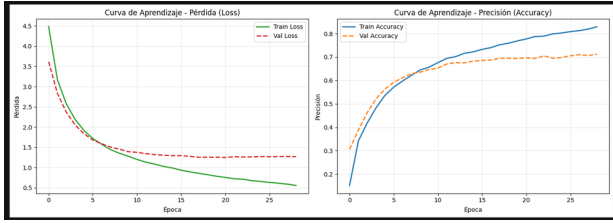
C. MLP TRAINING AND REGULARIZATION

The MLP was trained for up to 50 epochs with Early Stopping (patience = 8, monitoring validation loss). The best configuration (`hidden_dim=128`, `lr=0.001`) achieved the lowest validation loss of 1.2692. As shown in Fig. 5, training was halted before overfitting began, restoring the optimal epoch weights.

Table 2 summarizes the MLP hyperparameter search.

TABLE 2. MLP Hyperparameter Search Results (Validation Set)

Hidden Dim	LR	Best Val Loss	Best Val Acc
128	0.001	1.2692	0.7132
256	0.001	1.2477	0.7122
256	0.0001	1.7169	0.5861

**FIGURE 5.** Learning curves of the best MLP configuration (hidden_dim=128, lr=0.001). Early stopping halted training at epoch 29, restoring the weights from the best validation epoch.

D. FINAL RESULTS AND DISCUSSION

Table 3 presents the final metrics computed on the strictly isolated test set for all three models.

TABLE 3. Model Comparison on the Held-Out Test Set (20%)

Model	Macro F1	Weighted F1	Top-5 Acc
Random Forest	0.2916	0.5590	0.8610
XGBoost (GPU)	0.3348	0.6116	0.8638
MLP (PyTorch)	—	—	0.7132 [†]

[†] Val Accuracy (Top-1). F1 scores not computed for MLP.

The results demonstrate the effectiveness of the proposed methodology. XGBoost outperformed Random Forest across all metrics, achieving a Macro F1 of 0.3348, Weighted F1 of 0.6116, and Top-5 Accuracy of 86.38%. The relatively low Macro F1 across all models is expected given that 38% of the 526 DRG classes have fewer than 5 training examples, making prediction of rare classes inherently difficult.

The superiority of XGBoost and MLP over sequential architectures (CNN-1D, LSTM) lies in the nature of TF-IDF vectorization. By transforming the medical history into a global frequency matrix, the chronological order of codes is sacrificed, but a statistical representation of pathological co-occurrences is obtained. Dense architectures (MLP) and gradient-boosted trees (XGBoost) proved theoretically and empirically superior at isolating which specific medical terms trigger a given DRG category in high-dimensional sparse spaces.

E. FEATURE SELECTION JUSTIFICATION

Age and Sex are fundamental demographic filters for certain DRGs (e.g., obstetric groups are exclusive to females within specific age ranges), providing strong prior information for classification. The clinical codes (ICD-10) represent the core medical reason for hospitalization. The output is the GRD code, the standardized administrative classification target.

VII. CONCLUSIONS

This study successfully developed and compared predictive models for DRG classification using discharge data from Hospital El Pino. By implementing a multimodal TF-IDF pipeline with XGBoost (GPU-accelerated) and a regularized deep MLP, significant improvements over baseline Random Forest were achieved, particularly in managing the extreme class imbalance of 526 DRG categories dominated by obstetric cases.

The XGBoost model emerged as the best overall performer with a Top-5 Accuracy of 86.38%, demonstrating that the correct DRG appears among the top 5 model suggestions in more than 86% of patient cases—a clinically actionable result for assisting medical coders. The model meets the study objective of providing a technical foundation for automating hospital administrative processes.

VIII. LIMITATIONS AND FUTURE WORK

A. LIMITATIONS

- **Class Imbalance:** DRGs with fewer than 5 training examples remain difficult to predict accurately, keeping Macro F1 scores below 0.35.
- **Loss of Temporal Order:** TF-IDF vectorization discards the sequential order of diagnoses and procedures, which may contain clinically relevant patterns.
- **Static Feature Space:** The 1,000-feature TF-IDF cap may exclude relevant rare medical terms.

B. FUTURE WORK

- **SMOTE / Data Augmentation:** Implement synthetic oversampling for minority DRG classes to improve Macro F1.
- **ClinicalBERT:** Integrate pre-trained medical language models to extract semantic meaning from diagnosis descriptions.
- **RAG-Enhanced Chatbot:** Explore Retrieval-Augmented Generation to extend the Phase 2 chatbot beyond static knowledge injection.

IX. CHATBOT DEVELOPMENT: KONOHA-BOT

A. DOMAIN AND MOTIVATION

As part of Phase 2, a domain-specific conversational agent was developed. The selected domain is the Naruto manga universe, chosen for its rich, well-defined lore that enables rigorous testing of factual precision and domain boundary enforcement. The chatbot, named KONOHA-BOT, functions as an encyclopedia of the ninja world.

B. SYSTEM ARCHITECTURE

The system follows a three-layer architecture:

- 1) **Frontend:** A single-page web interface (HTML/CSS/JavaScript) that handles user interaction.
- 2) **Backend:** A lightweight Python server (Flask) that manages conversation history and routes requests to the LLM.

- 3) **Inference Layer:** LM Studio serving DeepSeek-R1-Distill-Llama-8B (Q4 K M quantization, 4.92 GB) locally via an OpenAI-compatible REST API on <http://localhost:1234>, running on an NVIDIA RTX 4060.

C. PROMPT ENGINEERING STRATEGY

The chatbot's intelligence is controlled entirely through prompt engineering, without any fine-tuning. The system prompt fulfills three functions: (1) role definition—the model is instructed to act as a Naruto encyclopedia; (2) knowledge injection—the full `knowledge_base.txt` is embedded into the system prompt at startup (context stuffing); (3) behavioral constraints—explicit rules restrict responses to the target domain and define a fixed refusal message for out-of-scope queries. Temperature was set to 0.3 to favor deterministic, factual responses.

D. KNOWLEDGE BASE

The knowledge base was manually curated to cover all 10 validation questions, organized into thematic sections: Team 7 roster, Gaara's forehead symbol, chakra elemental cycle, Izanagi conditions, Itachi's massacre, Mangekyō Sharingan variants, Konoha founders, Sage Mode independence from Kurama, and the Infinite Tsukuyomi mechanics. Additional sections cover the nine Tailed Beasts, the Akatsuki, ninja hierarchy, and Kekkei Genkai.

E. PERFORMANCE EVALUATION

Each of the 10 validation questions was posed under two conditions: (A) Baseline—raw LM Studio with no system prompt or knowledge base; (B) KONOHA-BOT—full system with injected knowledge base.

The results demonstrate clear improvement with prompt engineering and the knowledge base. The baseline produced correct answers in only 1 out of 10 questions (Q9, partially), and critically hallucinated a character from a different franchise in Q1. KONOHA-BOT answered all 10 questions correctly. Three key patterns emerged: (1) Hallucination suppression—the baseline fabricated plausible-sounding but incorrect lore; (2) Domain boundary enforcement—behavioral constraints successfully rejected the out-of-scope query in Q7; (3) Response depth—KONOHA-BOT provided more structured, accurate answers for complex multi-step questions such as Q4 and Q10. Table 4 presents the comparative evaluation.

F. LIMITATIONS AND FUTURE WORK

The primary limitation is the static knowledge base: any update requires manual editing and a server restart. For long conversations, the combined system prompt and knowledge base may approach the model's 10,240-token context window, causing truncation. Future work should implement Retrieval-Augmented Generation (RAG) to enable efficient querying of a much larger knowledge base without injecting it entirely on every call.

TABLE 4. Comparative Evaluation: Baseline Model vs. KONOHA-BOT across the 10 Validation Questions

Question	Baseline (No KB)	KONOHA-BOT (With KB)
Q1: Team 7 sensei	Incorrect. Named "Aizen" (a Bleach character), confusing franchises entirely.	Correct. Identified Kakashi Hatake, explaining his Sharingan transplant from Obito.
Q2: Gaara's symbol	Incorrect. Claimed it was the Akatsuki membership symbol, fabricating story details.	Correct. Identified kanji (Ai) ("love"), explaining Gaara's self-tattooing after years of rejection.
Q3: Lightning chakra	Incorrect. Invented pseudo-scientific rationale; relationships were vague and wrong.	Correct. Stated advantage vs. Earth (Doton), disadvantage vs. Wind (Futon).
Q4: Izanagi conditions	Incorrect. Listed generic conditions unrelated to lore; attributed technique to Gaara.	Correct. Listed the three canon conditions: active Sharingan, Uchiha DNA, and Sage of Six Paths chakra.
Q5: Itachi's massacre	Partial. Mentioned Akatsuki and chakra control; omitted the coup and Danzō's order.	Correct. Explained Danzō's order, the planned coup d'état, and Itachi's condition to spare Sasuke.
Q6: Mangekyō variants	Partial. Described "Rays of Light" invented by Itachi; omitted blindness and transplant mechanism.	Correct. Explained trauma activation, progressive blindness, unique jutsus, and the Eternal variant's vision restoration.
Q7: Ichiraku recipe (OOD)	Failed boundary test. Provided a full ramen recipe treating a fictional restaurant as real.	Passed. Correctly refused with the defined out-of-scope message.
Q8: Founders of Konoha	Incorrect. Named Itachi and Gaara as founders—neither had any role in Konoha's founding.	Correct. Identified Hashirama Senju and Madara Uchiha, explaining the Senju-Uchiha alliance.
Q9: Sage Mode without Kurama	Partial. Reached correct conclusion but based on generic reasoning, not the specific lore.	Correct. Confirmed independence via Mt. Myoboku training and cited Jiraiya as precedent.
Q10: Infinite Tsukuyomi	Incorrect. Described it as "dream manipulation chakra," omitting God Tree, Moon reflection, and chakra absorption.	Correct. Provided all 7 canonical steps from Rinnegan activation to chakra absorption.

CODE AVAILABILITY

The full source code, Jupyter notebooks, and experimental setups for both the DRG prediction pipeline and the KONOHA-BOT are publicly available in the following GitHub repository: https://github.com/kenjikaese/Machine_learning_project01.

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